Experimental Study of Damage Detection by Data-Driven Subspace Identification and Finite-Element Model Updating

Jian-Huang Weng[1]; Chin-Hsiung Loh[2]; and Jann N. Yang[3]

Abstract: The need to identify the physical properties of a structure given its force-response (input-output) relationship is driven primarily by the need to validate the approximate solution models, such as finite-element models. This paper proposes a method, which combines the structural system identification and model updating techniques, for the damage detection of a steel frame structure and a RC frame. The damage detection procedure consists of two steps: (1) identifying the system dynamic characteristics using the subspace identification (SI) technique from input/output measurements and (2) developing a damage assessment method for structural members (including joints) based on a progressive finite-element model updating and a large-scale optimization using a nonlinear least-square technique. The proposed method was verified through a shaking table experimental study using: (1) a 1/4-scale six-story steel frame structure by loosening the connection bolts for damage simulations and (2) a two-story RC frame subject to different levels of ground excitations back to back. As demonstrated by experimental results, the proposed damage detection method, based on the combination of SI technique and the model updating approach, is very effective for the damage assessment of frame structures. The method not only can detect the damage locations but also can quantify the damage severities.

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Introduction

The need to identify the physical properties of a structure given its force-response (input-output) relationship is driven primarily by the need to validate the approximate solution models, such as finite-element (FE) models. The selected FE models should be able to represent the real structure as accurately as possible. In general, a FE model is defined by a large number of model parameters (Young's modulus, moment of inertia, connection rigidity, etc.) and not the response at all the degrees of freedom in the model can be measured. Furthermore, a modal test of the target structure can provide only a small number of modal properties (natural frequencies, damping ratios, and mode shapes) for the modification of the modal properties of the FE model. Therefore, updating a FE model to obtain a reliable and accurate prediction for the structural response remains to be a major challenge to date. In this connection, reducing the order of the FE model and implementing the artificial boundary conditions can be a viable method to deal with this challenge.

There are two strategies in the FE model updating. One strategy is to modify the system matrices, such as the stiffness and damping matrices, based on the measured data (Baruch and Bar-Itzhack 1978; Berman and Nagy 1983). This direct method improves the updated model in reproducing more accurate dynamic modal responses of the structure. The second strategy is to select the geometric and material properties of the structural elements as the updating model parameters (Friswell and Mottershead 1995; Fritzen and Kiefer (1992); Link 1999). Then, these parameters are modified until the differences in dynamic characteristics between the FE model and the real structure are minimized. Methods in this strategy, which are based on the frequency domain data, have been developed mainly by vibration engineers. Furthermore, most model-based damage detection procedures in structural mechanics have been derived from the model updating and error localization methods (Mottershead and Friswell 1993; Lallement and Andriambololona 1991; Fritzen and Zhu 1991). The aforementioned second strategy is more desirable for the damage detection because the updated model parameters are directly related to the localization and severity of damages. Therefore, the second strategy will be used for the FE model updating in this study. Likewise, the estimations of mass and stiffness of the structural elements (e.g., beam, column) can be easily obtained from the design details. On the other hand, the mechanical behavior of the connections (joints) is difficult to evaluate although these connections are vulnerable during sever excitations, especially for RC structures. Furthermore, the stiffness of structural joints may significantly influence the dynamical behavior of the entire structure (Jaspart 2000; Nethercot 2000). Hence, the joints of members should be considered as semirigid joints (artificial boundary condition), and the rotational stiffness of a joint should be considered as an updating parameter. Then, the damage at a joint can be reflected by the reduction of its rotational stiffness.

An improved method is to determine the modal responses (such as mode shapes) of the FE model from the experimental

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results. In this connection, the system identification method plays an important role in the determination of the modal properties from the experimental data. Many system identification approaches to extract the dynamic features from the measured experimental data have been developed in the literature. Methods, such as the subspace identification (SI) algorithms, have been developed to identify the discrete-time state-space model for linear systems. Enhancement, overviews, and comparisons of SI methods were given in Van Overschee and De Moor (1996). Experimental demonstrations of this method were published, e.g., in Peeters and Roeck (1999), Mevel and Basseville (2003), Fang et al. (2008) etc.

The method proposed in this paper consists of two parts. The first part presents the proposed procedures based on the SI method to identify system modal properties from the input/output experimental data. In the second part, a FE model updating method is proposed to determine the most suitable model of the system. Based on the FE model updating, the stiffness factors (bending stiffness of beams, rotational stiffness of joints) of the flexibly connected beam element were identified using a large-scale optimization technique. Experimental tests on the shaking table for a six-story steel frame and a two-story RC frame have been conducted. Applications of the proposed method to detect: (1) the location of loosened bolts of the steel frame and (2) the damage pattern of the RC frame, are demonstrated by the experimental results.

Experimental Setup for Structural Damage Identification

Two types of structures have been constructed for the study of damage detection using the proposed FE model updating method: one is a six-story steel frame and the other is a two-story RC frame. Experimental setups are described in the following.

Six-Story Steel Frame with Loosened Bolts to Simulate Damages

A six-story steel frame is used to verify the applicability of the proposed damage identification technique. As shown in Fig. 1, the six-story scale-down structure consists of a single bay with a 1.0 m by 1.5-m floor area and a 1.0-m story height. The dimension and connecting information are as follows: column: 150 $\times$25 mm (rectangular section), beam: 50$\times$50$\times$5 mm (L-section); floor slab: 1,000$\times$1,500$\times$20 mm; base plate: 1,700 $\times$1,700$\times$20 mm; beam floor: welding connection; column floor: bolted connection; base column: bolted connection; bracing floor: bolted connection. The mass on each floor (lumped mass) is 862.85 kg. The structure is bolted on the shaking table (National Center for Research on Earthquake Engineering shaking table with dimension 5$\times$5 m). The undamaged test structure was tested first using the white noise excitation with a peak acceleration of 50 gal The time-domain excitation and response signals of each floor were measured. For this six-story steel frame, five cases of joint damage scenarios were artificially introduced (loosened bolts on different floors) as shown in Fig. 1:

- Damage Case 1: loosened bolts on 1st floor.
- Damage Case 2: loosened bolts on 1st and 4th floors.
- Damage Case 3: loosened bolts on 4th floor.
- Damage Case 4: loosened bolts on 4th and 6th floors.
- Damage Case 5: loosened bolts on 6th floor.

Seismic responses for each damage case were recorded to identify the locations of the loosened bolts and the rotational stiffness reduction of the joints. For the simulations of five aforementioned damage cases (loosened bolts), the El Centro earthquake ground motion, normalized to a peak ground acceleration (PGA) of 100 gal was used as the input excitation (along *Y*-direction). Only the measured acceleration responses from each floor will be used in the proposed method to identify model parameters. Based on the experimental results, the dynamic characteristics (natural frequencies and mode shapes) of the experimental structure will be identified first using the SI method. In this approach, the number of degrees of freedom of the test structure is equal to the number of response measurements, which is six. A brief description of the SI method is given in the following section.

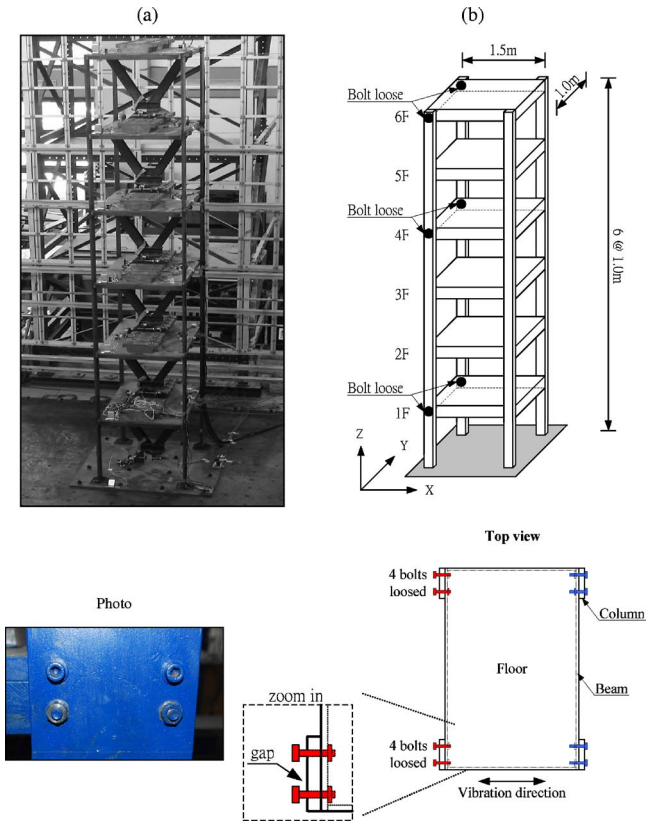


Fig. 1. Six-story steel frame for experimental study: (a) photo of the test structure; (b) a sketch showing the dimension of the structure and the prearranged possible damage locations (black dots indicating the tentative locations of loosened bolts for damage detection). The connection design between the floor and the column is also shown.

Two-Story RC Frame Subject to Different Levels of Ground Excitations

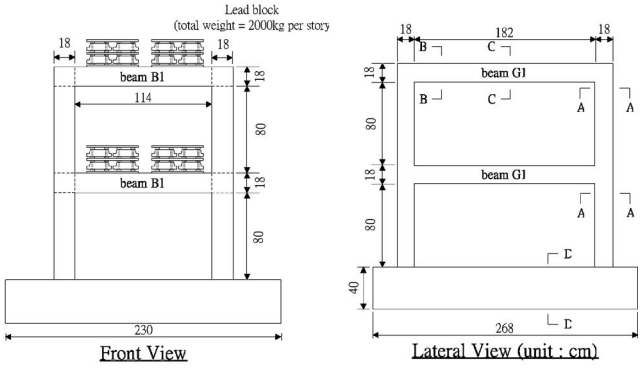
A 1/3-scale RC frame was designed following the ACI 318-05 design code and it was tested on the shaking table. The height of each floor is 0.98 m, and the plan dimension is 2.18 m by 1.5 m. The total weight per story is 2 t. The frame was designed with the first natural frequency of 12.0 Hz. Fig. 2 shows the dimension of the specimen. The structure was tested on the shaking table using various excitation intensities. The white noise and seismic excitations were applied back to back sequentially in a uniaxial direction. Only the acceleration response data of each floor were measured. Table 1 summarizes the excitation sequence of the test specimen on the shaking table.

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(a)



(b)

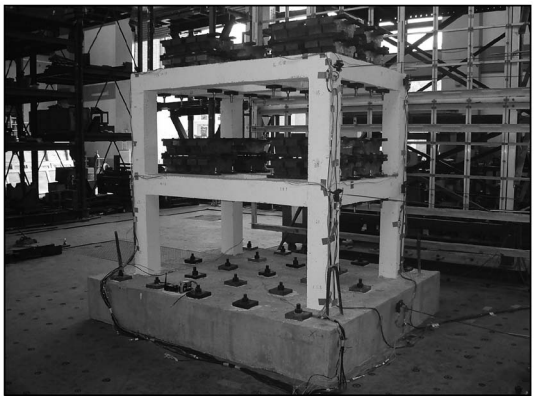


Fig. 2. Two-story RC frame: (a) dimension of the two-story RC frame (front view and lateral view); (b) test specimen on shaking table

Subspace Identification Method for State-Space Model

In this section the SI method is described briefly in the following. The SI method provides a systematic approach to determine the model order for a given accuracy and to derive the discrete state-space model. In contrast to the classical predictor error methods, the SI techniques derive models for linear systems solely by applying the well-conditioned operations, such as LQ and singular value decompositions (SVDs) of the block Hankel data matrices. Then, the system parameter matrices can be extracted from the block equation.

Consider a discrete state-space equation and a measurement (or observation) equation

$$\mathbf{x}_{k+1} = \mathbf{A}\mathbf{x}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k \tag{1a}$$

$$\mathbf{y}_k = \mathbf{C}\mathbf{x}_k + \mathbf{D}\mathbf{u}_k + \mathbf{v}_k \tag{1b}$$

where $\mathbf{A} \in \mathbf{R}^{n\times n}, \mathbf{B} \in \mathbf{R}^{n\times m}, \mathbf{C} \in \mathbf{R}^{l\times n}, \mathbf{D} \in \mathbf{R}^{l\times m}$=system matrix, control matrix, observer matrix, direct transmission matrix, respectively. In Eqs. (1*a*) and (1*b*), $\mathbf{A}$ is the discrete state matrix, $\mathbf{B}$ is the discrete input influence matrix, $\mathbf{u}_k \in \mathbf{R}^{m\times 1}$ and $\mathbf{y}_k \in \mathbf{R}^{l\times 1}$ are the input and output measurement vectors, respectively, $\mathbf{x}_k = \mathbf{x}(k\Delta t) \in \mathbf{R}^{n\times 1}$ is the discrete-time state vector, Δt is the sampling time, and $k \in n$. The discrete state-space model and output equations in Eqs. (1*a*) and (1*b*) include an additional system uncertainty process $\mathbf{w}_k$ and a measurement noise process $\mathbf{v}_k$, respectively.

Table 1. Back to Back Input Earthquake Amplitudes Applied to a Two-Story RC Frame

Excitation	Peak value
White noise 1	50gal
TCU078EW	200gal
White noise 2	50gal
TCU078EW	400gal
White noise 4	50gal
TCU078EW	600gal
White noise 5	50gal
TCU078EW	800gal
White noise 6	50gal
TCU078EW	1,000gal
White noises 7, 8	50gal
TCU078EW	1,200gal
White noise 9	50gal
TCU078EW	1,500gal
White noises 10, 11	50gal
TCU078EW	2,000gal
White noise 12	50gal
TCU078EW	1,200gal
White noise 13	50gal

Note: 1 g=1,000 gal.

Block Input-Output Equations

Let $\tilde{\mathbf{u}}$ be a matrix composed of a sequence of the input signal

$$\tilde{\mathbf{u}} = \begin{bmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_s \end{bmatrix} \in \mathbf{R}^{m\times s} \tag{2}$$

where s=total sampling number. Now, define the following block-Hankel matrix constructed from the input signal (Van Overschee and De Moor 1996):

$$\begin{bmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \cdots & \mathbf{u}_{j-1} \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_j \\ \cdots & \cdots & \cdots & \cdots \\ \mathbf{u}_{i-1} & \mathbf{u}_i & \cdots & \mathbf{u}_{i+j-2} \\ \hline \mathbf{u}_i & \mathbf{u}_{i+1} & \cdots & \mathbf{u}_{i+j-1} \\ \mathbf{u}_{i+1} & \mathbf{u}_{i+2} & \cdots & \mathbf{u}_{i+j} \\ \cdots & \cdots & \cdots & \cdots \\ \mathbf{u}_{2i-1} & \mathbf{u}_{2i} & \cdots & \mathbf{u}_{2i+j-2} \end{bmatrix} = \begin{bmatrix} \mathbf{U}_p \\ \mathbf{U}_f \end{bmatrix} = \begin{bmatrix} \mathbf{u}_0 & \mathbf{u}_1 & \cdots & \mathbf{u}_{j-1} \\ \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_j \\ \cdots & \cdots & \cdots & \cdots \\ \mathbf{u}_i & \mathbf{u}_{i+1} & \cdots & \mathbf{u}_{i+j-1} \\ \hline \mathbf{u}_{i+1} & \mathbf{u}_{i+2} & \cdots & \mathbf{u}_{i+j} \\ \mathbf{u}_{i+2} & \mathbf{u}_{i+3} & \cdots & \mathbf{u}_{i+j+1} \\ \cdots & \cdots & \cdots & \cdots \\ \mathbf{u}_{2i-1} & \mathbf{u}_{2i} & \cdots & \mathbf{u}_{2i+j-2} \end{bmatrix}$$
$$\equiv \begin{bmatrix} \mathbf{U}_p^+ \\ \mathbf{U}_f^- \end{bmatrix} \in \mathbf{R}^{2mi\times j} \tag{3}$$

where i=number of block rows defined by the user and it must be larger than the order n of the system; j=number of block columns; and $j=s-2i+2$ if all samples are used. The input block-Hankel matrix can be divided into two parts: the past part $\mathbf{U}_p \in \mathbf{R}^{mi\times j}$ and the future part $\mathbf{U}_f \in \mathbf{R}^{mi\times j}$. Besides, $\mathbf{U}_p^+ \in \mathbf{R}^{m(i+1)\times j}$ and $\mathbf{U}_f^- \in \mathbf{R}^{m(i-1)\times j}$ are defined by shifting the border between $\mathbf{U}_p$ and $\mathbf{U}_f$ one block row down. Similarly, the state space of the system can also be represented by the past and the future parts as follows:

$$\mathbf{X}_p \equiv \begin{bmatrix} \mathbf{x}_0 & \mathbf{x}_1 & \cdots & \mathbf{x}_{j-1} \end{bmatrix} \in \mathbf{R}^{n\times j} \quad \text{and} \quad \mathbf{X}_f = \begin{bmatrix} \mathbf{x}_i & \mathbf{x}_{i+1} & \cdots & \mathbf{x}_{i+j-1} \end{bmatrix} \in \mathbf{R}^{n\times j} \tag{4}$$

The output of the system can also be expressed in a similar form. Furthermore, two special Hankel matrices consisting of both inputs and outputs can also be defined as

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$$\mathbf{Z}_p \equiv \begin{bmatrix} \mathbf{U}_p \\ \mathbf{Y}_p \end{bmatrix} \in \mathbf{R}^{(m+l)i\times j} \quad \text{and} \quad \mathbf{Z}_p^+ \equiv \begin{bmatrix} \mathbf{U}_p^+ \\ \mathbf{Y}_p^+ \end{bmatrix} \in \mathbf{R}^{(m+l+2)i\times j} \tag{5}$$

After arranging the input $\mathbf{u}_k$, output $\mathbf{y}_k$ and state $\mathbf{x}_k$ in the block-Hankel matrix, the input-output matrix can be expressed as

$$\mathbf{Y}_p = \mathbf{\Gamma}_i \mathbf{X}_p + \mathbf{H}_i \mathbf{U}_p + \boldsymbol{\nu}_p \tag{6a}$$

$$\mathbf{Y}_f = \mathbf{\Gamma}_i \mathbf{X}_f + \mathbf{H}_i \mathbf{U}_f + \boldsymbol{\nu}_f \tag{6b}$$

$\mathbf{H}_i$=block Toeplitz matrix of the impulse response from u to y and $\boldsymbol{\nu}_p$, and where $\boldsymbol{\nu}_p$=measurement noise and $\mathbf{\Gamma}_i$=observability matrix (the subscript i has been defined previously) given by

$$\mathbf{\Gamma}_i \equiv \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \mathbf{CA}^2 \\ \cdots \\ \mathbf{CA}^{i-1} \end{bmatrix} \in \mathbf{R}^{li\times n} \tag{7}$$

and $\mathbf{H}_i^d$ is the lower block triangular Toeplitz matrix defined as

$$\mathbf{H}_i^d \equiv \begin{bmatrix} \mathbf{D} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{CB} & \mathbf{D} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{CAB} & \mathbf{CB} & \mathbf{D} & \cdots & \mathbf{0} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ \mathbf{CA}^{i-2}\mathbf{B} & \mathbf{CA}^{i-3}\mathbf{B} & \mathbf{CA}^{i-4}\mathbf{B} & \cdots & \mathbf{D} \end{bmatrix} \in \mathbf{R}^{li\times mi} \tag{8}$$

The input/output state space time-domain realization method can be used to find the system matrices $\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}$ from a given set of input and output data.

The SI algorithm is developed based on the following geometric concept. A row vector of a block-Hankel matrix is considered as a vector in the j-dimensional ambient space (where j is the number of block columns). From the projection theorem, the oblique projection matrix $\mathbf{O}_i$ is equal to the product of the observability matrix $\mathbf{\Gamma}_i$ and the estimated Kalman filter state sequence, $\widetilde{\mathbf{X}}_i \in \mathbf{R}^{n\times j}$, which can be described as follows:

$$\mathbf{Y}_f /_{\mathbf{U}_f} \mathbf{Z}_p = \mathbf{O}_i = \mathbf{\Gamma}_i \widetilde{\mathbf{X}}_i \in \mathbf{R}^{li\times j} \tag{9}$$

Detail verification of this theorem can be found in Van Overschee's paper (Van Overschee and De Moor 1996). Similarly, a shifted oblique projection is equal to

$$\mathbf{Y}_f^- /_{\mathbf{U}_f^-} \mathbf{Z}_p^+ = \mathbf{O}_{i+1} = \mathbf{\Gamma}_{i-1} \widetilde{\mathbf{X}}_{i+1} \in \mathbf{R}^{l(i-1)\times j} \tag{10}$$

Once $\mathbf{O}_i$ and $\mathbf{O}_{i+1}$ are computed from the oblique projections of input/output data, the system parameters can be determined by the following procedures:

1. Calculate the SVD of $\mathbf{O}_i$ and determine the order n of the system by retaining larger singular values in $\mathbf{S}$

$$\mathbf{O}_i = \mathbf{USV}^T = (\mathbf{U}_1\mathbf{U}_2)\begin{pmatrix} \mathbf{S}_1 & 0 \\ 0 & \$_2 \end{pmatrix}\begin{pmatrix} \mathbf{V}_1^T \\ \mathbf{V}_2^T \end{pmatrix} \approx \mathbf{U}_1\mathbf{S}_1\mathbf{V}_1^T \tag{11}$$

where

$$\text{diag}(\mathbf{S}_1) = \begin{bmatrix} s_1 & s_2 & \ldots & s_n \end{bmatrix}^T$$

$$\text{diag}(\mathbf{S}_2) = \begin{bmatrix} s_{n+1} & s_{n+2} & \ldots & s_{li} \end{bmatrix}^T$$

and S_2 indicates the smaller eigenvalues which can ignored.

2. Calculate the observability matrix $\mathbf{\Gamma}_i$

$$\mathbf{\Gamma}_i = \mathbf{U}_1\mathbf{S}_1^{1/2} \tag{12}$$

3. Repeat Steps 1 and 2 and use $\mathbf{O}_{i+1}$ to calculate $\mathbf{\Gamma}_{i-1}$ (hold n in Step 1).
4. Calculate the estimated Kalman filter state sequences $\widetilde{\mathbf{X}}_i$ and $\widetilde{\mathbf{X}}_{i+1}$

$$\widetilde{\mathbf{X}}_i = \mathbf{\Gamma}_i^{\dagger}\mathbf{O}_i \tag{13}$$

$$\widetilde{\mathbf{X}}_{i+1} = \mathbf{\Gamma}_{i-1}^{\dagger}\mathbf{O}_{i+1} \tag{14}$$

5. Solve the following set of linear equations for $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$, and $\mathbf{D}$

$$\begin{bmatrix} \widetilde{\mathbf{X}}_{i+1} \\ \mathbf{Y}_{i|i} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{bmatrix}\begin{bmatrix} \widetilde{\mathbf{X}}_i \\ \mathbf{U}_{i|i} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\rho}_w \\ \boldsymbol{\rho}_v \end{bmatrix} \tag{15}$$

where

$$\mathbf{Y}_{i|i} = \begin{bmatrix} \mathbf{y} & \mathbf{y}_{i+1} & \cdots & \mathbf{y}_{i+j-1} \end{bmatrix} \in \mathbf{R}^{l\times j}$$

and

$$\mathbf{U}_{i|i} = \begin{bmatrix} \mathbf{u}_i & \mathbf{u}_{i+1} & \cdots & \mathbf{u}_{i+j-1} \end{bmatrix} \in \mathbf{R}^{m\times j}$$

6. Calculate the eigenvalue λ_k and eigenvector $\psi_k \in \mathbf{R}^{n\times 1}$ of $\mathbf{A}$

$$\det(\mathbf{A} - \lambda_k\mathbf{I}) = 0 \tag{16a}$$

$$(\mathbf{A} - \lambda_k\mathbf{I})\psi_k = 0 \tag{16b}$$

It should be noticed that the eigenvalues of $\mathbf{A}$ occur in complex conjugated pairs and the subscript "k" denotes the number of these pairs.

7. Determine the frequency ω_k and damping coefficient ξ_k from λ_k

$$\omega_k = \frac{a_k}{2\pi\Delta t}(\text{rad/s}) \quad \text{and} \quad \xi_k = \frac{|b_k|}{\sqrt{a_k^2 + b_k^2}} \tag{17}$$

where

$$a_k = \left|\arctan\left[\frac{\text{Im}(\lambda_k)}{\text{Re}(\lambda_k)}\right]\right| \quad \text{and} \quad b_k = \ln(\lambda_k)$$

8. Determine the mode shape $\mathbf{\Psi}_k$ (with corresponding frequency ω_k) from $\mathbf{C}$ and ψ_k

$$\mathbf{\Psi}_k = \mathbf{C}\psi_k \tag{18}$$

Based on the aforementioned method, the system matrices $(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})$ as well as the modal frequencies and mode shapes of the six-story steel frame can be identified from the input/output measurements. The identified first five modal frequencies of the undamaged structure are 1.05, 3.50, 6.12, 8.98, and 11.91 Hz, and the corresponding modal damping ratios are 1.14, 0.62, 0.52, 0.40, and 0.31%, respectively. Similar identification approach has been applied to all damage cases and the results are shown in Table 2.

Method for Distinguishing True Modes from Noise Modes

Based on the discrete state-space model, the distinction between the structural modes and the spurious modes can be made using the stability diagrams. As the true system order is often unknown, a practice in the modal analysis is to calculate the modal parameters for increasing order n (Peeters and Roeck 1999). If n is

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Table 2. Identified Natural Frequencies and Damping Ratios for Six Different Experimental Cases Using SI Technique

Experiment case	System properties identified by SSI	Mode 1	2	3	4	5	6
Undamaged	Frequency (Hz)	1.05	3.50	6.12	8.98	11.91	14.19
	Damping ratio (%)	1.14	0.62	0.52	0.40	0.31	0.25
1F bolt loosened	Frequency (Hz)	0.94	3.33	6.06	8.99	11.90	14.14
	Damping ratio (%)	1.37	0.70	0.56	0.46	0.33	0.24
1F and 4F bolt loosened	Frequency (Hz)	0.91	2.95	6.04	8.61	11.73	14.09
	Damping ratio (%)	1.28	0.68	0.51	0.42	0.31	0.26
4F bolt loosened	Frequency (Hz)	1.00	3.15	6.14	8.63	11.79	14.15
	Damping ratio (%)	1.28	0.66	0.55	0.42	0.28	0.26
4F and 6F bolt loosened	Frequency (Hz)	1.01	3.14	5.81	8.22	11.53	14.02
	Damping ratio (%)	1.29	0.78	0.64	0.47	0.32	0.27
6F bolt loosened	Frequency (Hz)	1.05	3.43	5.88	8.61	11.63	14.09
	Damping ratio (%)	1.23	0.63	0.59	0.57	0.40	0.25

higher than the true system order, the noise mode will be involved, but the mathematical poles thus obtained are different for different model if the noise is purely white. Hence, the poles of physical system can be detected by comparing the modal parameters for different model orders. To identify a suitable system order and to separate the true modes from the noise modes, for each specified model order n, the following three evaluation steps are proposed: (1) SVD; (2) output modal accuracy correlation (OMAC); and (3) weighted phase error (WPE).

To distinguish the true modes from the noise modes, we first determine the system order n based on the SVD of $\mathbf{O}_i$ [see Eq. (11)]. For example, for the six-story steel frame considered in this study, the singular value is defined as

$$C_r^{\mathrm{SVD}} = \frac{s_1}{1{,}000} \Rightarrow \begin{cases} s_k \geq C_r^{\mathrm{SVD}} & \text{for } k = 1 \sim n \\ s_k < C_r^{\mathrm{SVD}} & \text{for } k = n+1 \sim li \end{cases} \tag{19}$$

The system order n can be determined from the singular value greater than the assign value ($s_1/1{,}000$).

After the n-value is determined using SVD, the OMAC is then applied. Let $\mathbf{A}_m$ and $\mathbf{C}_m$ represent sequentially the system matrix $\mathbf{A}$ and the observer matrix $\mathbf{C}$ in the modal coordinate. The eigensolution of the system matrix $\mathbf{A}$ forms: (1) a diagonal matrix $\mathbf{A}_m$ of eigenvalues; and (2) a full matrix $\phi \in \mathbf{R}^{n\times n}$ whose columns are the corresponding eigenvectors, i.e., $\phi^{-1}\mathbf{A}\phi = \mathbf{A}_m$, as shown in the following equations:

$$\mathbf{A}_m = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix} \in \mathbf{R}^{n\times n} \quad \text{and} \quad \mathbf{C}_m = \mathbf{C}\phi$$
$$= [\boldsymbol{\Psi}_1 \boldsymbol{\Psi}_2 \ldots \boldsymbol{\Psi}_n] \in \mathbf{R}^{l\times n} \tag{20}$$

Using the modal coordinate, one can reconstruct the modal observability matrix from the identified model as follows:

$$\overline{\boldsymbol{\Gamma}}_{m,i} = \begin{bmatrix} \mathbf{C}_m \\ \mathbf{C}_m\mathbf{A}_m \\ \vdots \\ \mathbf{C}_m\mathbf{A}_m^{i-1} \end{bmatrix} = \begin{bmatrix} \begin{pmatrix} \boldsymbol{\Psi}_1 \\ \boldsymbol{\Psi}_1\lambda_1 \\ \vdots \\ \boldsymbol{\Psi}_1\lambda_1^{i-1} \end{pmatrix} & \begin{pmatrix} \boldsymbol{\Psi}_2 \\ \boldsymbol{\Psi}_2\lambda_2 \\ \vdots \\ \boldsymbol{\Psi}_2\lambda_2^{i-1} \end{pmatrix} & \cdots & \begin{pmatrix} \boldsymbol{\Psi}_n \\ \boldsymbol{\Psi}_n\lambda_n \\ \vdots \\ \boldsymbol{\Psi}_n\lambda_n^{i-1} \end{pmatrix} \end{bmatrix}$$
$$= [\overline{\boldsymbol{\varphi}}_1 \overline{\boldsymbol{\varphi}}_2 \ldots \overline{\boldsymbol{\varphi}}_n] \in \mathbf{R}^{li\times n} \tag{21}$$

Different from the modal analysis, the modal observability matrix can also be obtained directly from the data decomposition [see Eq. (12) of the SI method] as follows:

$$\tilde{\boldsymbol{\Gamma}}_{m,i} = \boldsymbol{\Gamma}_i\phi = \mathbf{U}_1\mathbf{S}_1^{1/2}\phi = [\tilde{\boldsymbol{\varphi}}_1 \tilde{\boldsymbol{\varphi}}_2 \ldots \tilde{\boldsymbol{\varphi}}_n] \in \mathbf{R}^{li\times n} \tag{22}$$

The OMAC is defined as the coherence between the vectors $\bar{\varphi}_k$ in Eq. (21) and $\tilde{\varphi}_k$ in Eq. (22)

$$\mathrm{OMAC}_k = \frac{|\overline{\boldsymbol{\varphi}}_k^* \tilde{\boldsymbol{\varphi}}_k|}{\sqrt{\overline{\boldsymbol{\varphi}}_k^* \overline{\boldsymbol{\varphi}}_k}\sqrt{\tilde{\boldsymbol{\varphi}}_k^* \tilde{\boldsymbol{\varphi}}_k}} (k = 1 \sim n) \tag{23}$$

where the superscript " *" denotes the Hermitian (complex conjugate) transpose. A larger value of OMAC indicates a higher correlation. In this study, a criterion is established by defining the OMAC value to be larger than a specified value, e.g., $\mathrm{OMAC}_k \geq C_r^{\mathrm{OMAC}} = 0.999$.

Following the SVD and OMAC approaches, the third step is to examine the phase of the identified modes. Let $\boldsymbol{\Psi}_k$ be the complex vector of the kth mode shape of the system. The normalized magnitude of $\boldsymbol{\Psi}_k$ is defined as

$$\frac{\mathrm{abs}(\boldsymbol{\Psi}_k)}{\mathrm{length}[\mathrm{abs}(\boldsymbol{\Psi}_k)]} = \begin{bmatrix} \alpha_{k1} \\ \alpha_{k2} \\ \vdots \\ \alpha_{kl} \end{bmatrix} \in \mathbf{R}^{l\times 1} \quad (0 \leq \alpha \leq 1) \tag{24}$$

where abs(.)=magnitude of a complex vector inside the parentheses and length(.)=length of the vector inside the parentheses. The phase of $\boldsymbol{\Psi}_k$ is defined as

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$$\text{phase}(\Psi_k) = \begin{bmatrix} \beta_{k1} \\ \beta_{k2} \\ \vdots \\ \beta_{kl} \end{bmatrix} \in \mathbf{R}^{l\times 1} \quad (-\pi \le \beta < \pi) \tag{25}$$

where phase(.) represents the phase (in radians) of the complex vector inside the parentheses. For a particular kth mode, the phase difference between two different measurement locations should be 0 or π, i.e.

$$|\beta_{kq} - \beta_{kr}| = 0 \quad \text{or} \quad \pi \quad (q \neq r) \tag{26}$$

Therefore, the $\text{WPE}_k(\gamma)$ for the kth mode is defined as the determination of γ value with the magnitude of the mode shape as weighting factor. This calculation runs for each indentified mode ($k=1\sim n$), i.e.

$$\text{WPE}_k(\gamma) = \begin{bmatrix} \alpha_{k1} & \alpha_{k2} & \dots & \alpha_{kl} \end{bmatrix} \cdot \begin{bmatrix} \min(|\beta_{k1}-\gamma|, |\beta_{k1}-\gamma+\pi|) \\ \min(|\beta_{k2}-\gamma|, |\beta_{k2}-\gamma+\pi|) \\ \vdots \\ \min(|\beta_{kl}-\gamma|, |\beta_{kl}-\gamma+\pi|) \end{bmatrix} \quad (0 \le \gamma < \pi) \tag{27}$$

The kth mode can be selected from the minimum value of $\text{WPE}_k(\gamma)$ and the criterion for the WPE in this study is selected to be $\text{WPE}_{k,\min} \le C_r^{\text{WPE}} = 0.05$.

All the preceding procedures can be performed visually in a so-called stabilization diagram, in which the eigenfrequencies corresponding to the system poles are plotted in increasing system orders. Through the use of SVD, OMAC, and WPE procedures sequentially, the true modes can be distinguished from the noise modes.

Finite-Element-Based Model Updating for Damage Detection

The goal of model updating is to find a set of structural model parameters that produces the dynamic properties of the FE model closer to those of the experimental results. In practice, it is not easy to select a set of appropriate model parameters. The discrepancies between the modal properties of both FE and the experimental models are due to a reduced number of parameters. Hence, the selection of suitable parameters is the key to success in the model updating. For the detection of structural damage using an iterative FE model updating, it is necessary to develop a progressive approach to update the model parameters in order to obtain the true damage condition. For the progressive approach in model updating, three sequential steps are proposed in the following. The first step is to define a FE model, which can represent the true structural system, and to match the true boundary conditions as close as possible. In this paper, two different types of connection are considered: (1) the beam-column connection element; and (2) the column-support connection element. The second step is to verify the model with the experimental data and to validate the system modal characteristics (such as modal frequencies and damping ratios). The third step is to update the FE model by introducing more model parameters in order to reflect the true structural behavior. This can be accomplished by minimizing an objective function, e.g., sum-square errors, in the time domain.

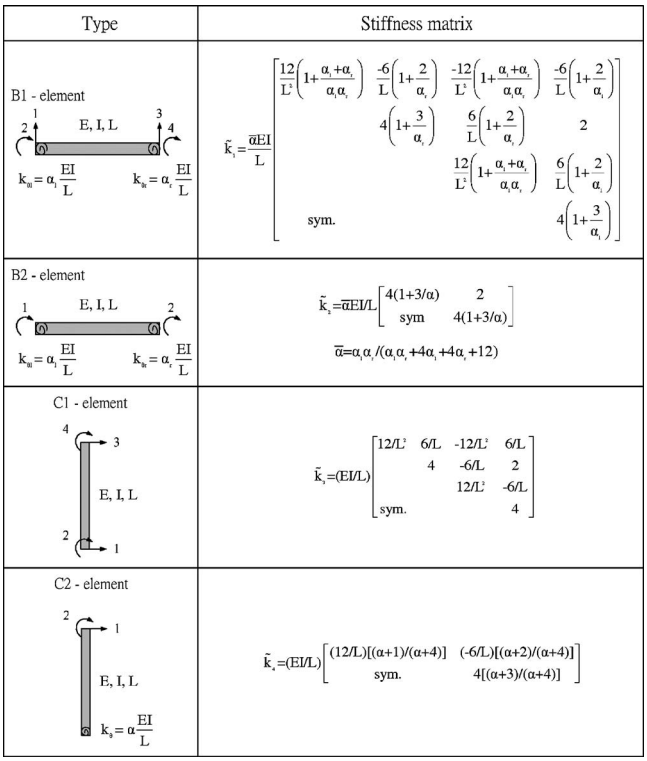


Fig. 3. Different types of beam elements and their corresponding element stiffness matrices

Damage Detection of a Six-Story Steel Frame

Modeling of Beam/Column Connections by Semirigid Joints

The conventional approach for structural analyses assumes that the joints are either pinned or completely rigid. Since the joint behavior has an important contribution to the response of the entire structure, the performance of the joints has been a critical factor in many structural collapses. Among possible modes of deformation in the semirigid connections, the most important mode is the rotational deformation caused by a bending moment. The flexural behavior of a FE may be represented by the element flexural rigidity and the joint rotational rigidity. The joint damage may be quantified by the reduction of the rotational rigidity. The semirigid connection is modeled in this paper as a zero-length rotational spring at both ends of a beam, for example, at the bottom of the first floor column or at both ends of the floor beam. The rotational rigidity (rotational stiffness) of a beam-to-column connection k_θ is express as

$$k_\theta = \alpha \frac{EI}{L} \tag{28}$$

where α = unknown multiplication factor. If the stiffness can be evaluated before and after the damage, then the damage severity can be determined from their reductions. In this experimental study, four different beam elements are used to form the system stiffness matrix (B1: beam element with joint rotation and translation at both ends; C1: column element; C2: column element with joint rotational stiffness at one end; B2: beam element with joint rotation at both ends). Fig. 3 shows four different types of elements used for this experimental study. After updating the model parameters, a total of nineteen unknown multiplication factors will be identified as indicated in Table 3. The vector of these parameters is defined as

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Table 3. Selected Model Parameters for the Test Structure (Six-Story Steel Frame)

Model parameter	Description
α_1	Rotational stiffness of the column spring: $\hat{k}_{\theta,c}=\alpha_1\cdot(E\hat{I}_{c,1}/L_c)$
$\alpha_2\sim\alpha_7$	Rotational stiffness of the beam spring: $\hat{k}_{\theta,b_i}=\alpha_{i+1}\cdot(E\hat{I}_{b,i}/L_b)$ $(i=1\sim6)$
$\alpha_8\sim\alpha_{13}$	Moment of inertia of column section: $\hat{I}_{c,i}=\alpha_{i+7}I_c$ $(i=1\sim6)$
$\alpha_{14}\sim\alpha_{19}$	Moment of inertia of beam section: $\hat{I}_{b,i}=\alpha_{i+13}I_b$ $(i=1\sim6)$

Note: E, I_c, and I_b are estimated from the design details.

$$\boldsymbol{\beta}_I=\langle\alpha_1\quad\alpha_2\quad\cdots\quad\alpha_{19}\rangle \tag{29}$$

Since members of the experimental frame are steel strips, the shear deformations for this type of sections are negligible when comparing with the flexural ones. Fig. 4 shows the FE model of the six-story steel frame with a total of 19 parameters to be updated.

Progressive Updating Procedures

From the physical point of view, the model updating technique is to update the model parameters to reflect the true structural behavior. To identify the model parameters, we minimize an objective function that represents the discrepancy between the FE model and the real structure. In this section, two approaches are proposed to estimate the model parameters. In the first approach, the objective function is defined as the quadratic sum-square discrepancies between the modal properties of the undamped FE model and the experimental results:

$$J_I=\sum_{k=1}^{6}\mathbf{e}_k^2(\hat{\boldsymbol{\beta}}_I) \tag{30}$$

where

$$\mathbf{e}_k(\hat{\boldsymbol{\beta}}_I)=\hat{\mathbf{K}}(\hat{\boldsymbol{\beta}}_I)\boldsymbol{\Psi}_k^{SI}-\omega_k^{SI}\hat{\mathbf{M}}\boldsymbol{\Psi}_k^{SI} \tag{31}$$

In Eq. (31), $\boldsymbol{\Psi}_k^{SI}$ and $\omega_j^{SI}=k$th mode shape and modal frequency of the structure, which have been identified from the experimental data using the SI technique already. To identify the system natural frequency and damping ratio, the stability diagram can be used. Fig. 5 shows the results of system poles using the SVD, OMAC, and WPE methods. Fig. 5(a) indicates the plot of the singular value with respect to the number of the singular value for different number of block rows, whereas Fig. 5(b) shows the stable poles with $C_r^{\mathrm{SVD}}=S_1/1{,}000$. Fig. 5(c) presents the poles with $C_r^{\mathrm{SVD}}=s_1/1{,}000$ and $\mathrm{OMAC}_k\geq C_r^{\mathrm{OMAC}}=0.999$, and Fig. 5(d)

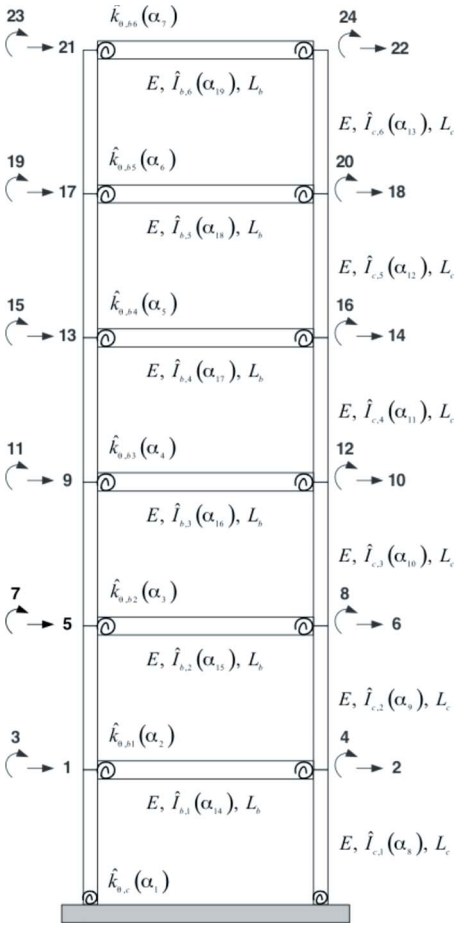


Fig. 4. FE model of the six-story steel frame and 19 updated parameters associated with structural elements

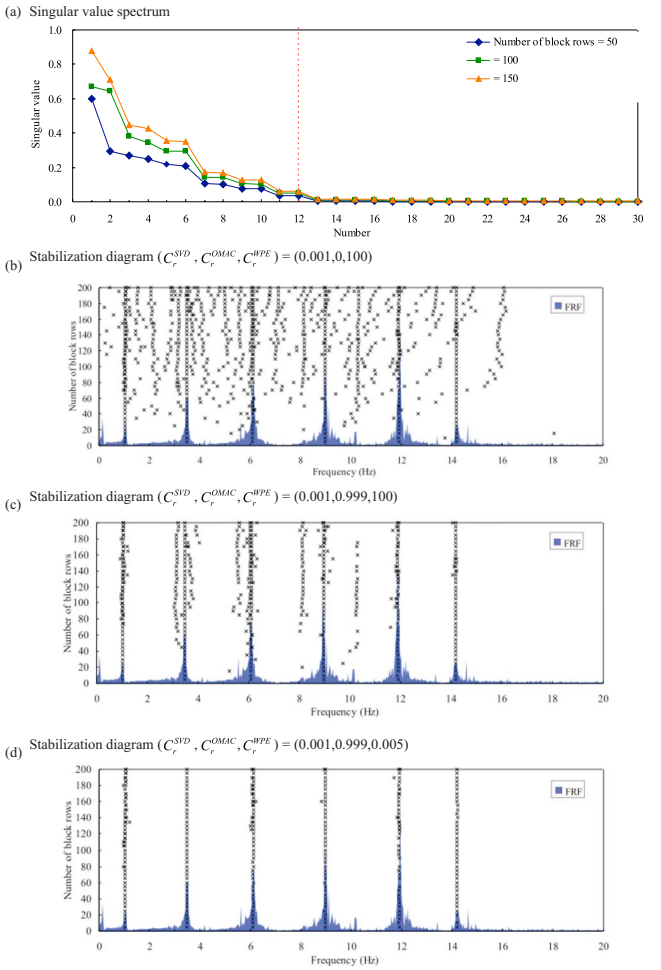


Fig. 5. Stabilization diagram for the six-story steel frame: (a) plot of singular value with respect to the number of the singular value for different number of block rows; (b) stable poles with $C_r^{\mathrm{SVD}}=s_1/1{,}000$; (c) poles with $C_r^{\mathrm{SVD}}=s_1/1{,}000$; and $\mathrm{OMAC}_k\geq C_r^{\mathrm{OMAC}}=0.999$; and (d) poles with $C_r^{\mathrm{SVD}}=s_1/1{,}000$, $\mathrm{OMAC}_k\geq C_r^{\mathrm{OMAC}}=0.999$, and $\mathrm{WPE}_{k,\min}\leq C_r^{\mathrm{WPE}}=0.05$

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Table 4. Updated Model Parameters Associated with the Structural Element for Each Updated Step

Associated structural element		Step 1	Step 2	Step 3	Step 4	Step 5
Rotational spring in column		α_1	α_1	α_1	α_1	α_1
Rotational spring in beam	1F	α_2	α_2	α_2	α_2	α_2
	2F	α_2	α_3	α_3	α_3	α_3
	3F	α_2	α_4	α_4	α_4	α_4
	4F	α_2	α_5	α_5	α_5	α_5
	5F	α_2	α_6	α_6	α_6	α_6
	6F	α_2	α_7	α_7	α_7	α_7
Column moment of inertia	1F	—	—	α_8	α_8	α_8
	2F	—	—	α_8	α_9	α_9
	3F	—	—	α_8	α_{10}	α_{10}
	4F	—	—	α_8	α_{11}	α_{11}
	5F	—	—	α_8	α_{12}	α_{12}
	6F	—	—	α_8	α_{13}	α_{13}
Beam moment of inertia	1F	—	—	α_9	α_{14}	α_{14}
	2F	—	—	α_9	α_{15}	α_{15}
	3F	—	—	α_9	α_{16}	α_{16}
	4F	—	—	α_9	α_{17}	α_{17}
	5F	—	—	α_9	α_{18}	α_{18}
	6F	—	—	α_9	α_{19}	α_{19}
Damping ratio	Mode1	—	—	—	—	α_{20}
	Mode2	—	—	—	—	α_{21}
	Mode3	—	—	—	—	α_{22}
	Mode4	—	—	—	—	α_{23}
	Mode5	—	—	—	—	α_{24}
	Mode6	—	—	—	—	α_{25}

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shows the poles with $C_r^{\mathrm{SVD}}=s_1/1{,}000$, $\mathrm{OMAC}_k \geq C_r^{\mathrm{OMAC}}=0.999$, and $\mathrm{WPE}_{k,\min} \leq C_r^{\mathrm{WPE}}=0.05$. From Fig. 5(d) the system natural frequencies can be clearly identified.

In Eq. (31), $\hat{\mathbf{K}}(\hat{\boldsymbol{\beta}}_I)$ is the stiffness matrix, which is a function of unknown model parameters $\hat{\boldsymbol{\beta}}_I$. Because only the floor acceleration responses are measured during the experiment, a total of six modes are obtained from the experimental results (i.e., $k=1 \sim 6$). Note that $\hat{\mathbf{K}}(\hat{\boldsymbol{\beta}}_I)$ is a (6×6) stiffness matrix obtained by a static condensation from the stiffness matrix of the FE model. Since the objective function is a nonlinear function of the unknown parametric vector $\boldsymbol{\beta}_I$, the solution can be obtained using an iterative procedure. In this paper a new algorithm, referred to as the interior-reflective Newton approach, is used. This algorithm can be used for large-scale and sparse problems for minimizing a smooth nonlinear function involving many variables and subject to the constraints of upper and/or lower bounds for some variables (Coleman 1994, 1996). This interior-reflective Newton approach is globally and quadratic convergent.

Based on the identified model parameters $\boldsymbol{\beta}_I$ from the first approach, a second updating procedure will be used by increasing the model parameters, including the modal damping ratios. The vector of these parameters is now updated and defined as

$$\boldsymbol{\beta}_{II} = \langle \alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_{19} \quad \xi_1 \quad \xi_2 \quad \cdots \quad \xi_6 \rangle \tag{32}$$

where $\xi_1, \xi_2, \cdots, \xi_6$=modal damping ratios for the six modes. The new objective function is the sum-square differences between the predict acceleration responses $\hat{\mathbf{y}}_k(\hat{\boldsymbol{\beta}}_{\mathrm{II}}, t)$ and the measured acceleration response data $\mathbf{y}_k(t)$ in the time domain

$$J_{\mathrm{II}} = \sum_{k=1}^{6} \int \{\mathbf{y}_k(t) - \hat{\mathbf{y}}_k(\hat{\boldsymbol{\beta}}_{\mathrm{II}}, t)\}^2 dt \tag{33}$$

For estimating the predicted responses, the proportional damping is assumed and the system damping matrix is expressed as

$$\hat{\mathbf{C}}(\hat{\boldsymbol{\beta}}_{\mathrm{II}}) = (\boldsymbol{\Phi}^T)^{-1} \hat{\mathbf{C}}_d(\hat{\boldsymbol{\beta}}_{\mathrm{II}}) \boldsymbol{\Phi}^{-1} \tag{34}$$

where

$$\hat{\mathbf{C}}_d(\hat{\boldsymbol{\beta}}_{\mathrm{II}}) = \begin{bmatrix} 2\hat{m}_1\hat{\omega}_1\hat{\xi}_1 & 0 & 0 & 0 \\ 0 & 2\hat{m}_2\hat{\omega}_2\hat{\xi}_2 & 0 & 0 \\ 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & 2\hat{m}_6\hat{\omega}_6\hat{\xi}_6 \end{bmatrix} \quad \text{and}$$

$$\boldsymbol{\Phi} = \begin{bmatrix} \boldsymbol{\Psi}_1^{SI} & \boldsymbol{\Psi}_2^{SI} & \cdots & \boldsymbol{\Psi}_6^{SI} \end{bmatrix}$$

In Eq. (34), the mode shape $\boldsymbol{\Phi}$ has been estimated from the SI method already.

Results of Model Updating

Based on the preceding progressive updating method, the parametric updating is proposed to be carried out in five steps for this particular structure. For each step, a set of model parameters will be assigned. Through updating model parameters, the FE model becomes more realistic and less physical constraints. These five steps of updating procedures are shown in Table 4 and described in the following:

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Table 5. Comparison on Modal Frequencies Estimated from SI Method, Initial FEM, Step-1 to Step-5 Model Updating Method (Undamaged Structure)

Identification stage		Modal frequency					
		Mode1	Mode2	Mode3	Mode4	Mode5	Mode6
SI		1.04	3.48	6.11	8.96	11.90	14.18
Initial	Evaluation (Hz)	1.71	5.07	8.23	11.03	13.26	14.70
Step1	Evaluation (Hz)	1.10	3.43	6.05	8.97	11.93	14.28
Step2	Evaluation (Hz)	0.92	3.19	5.96	8.94	11.89	14.23
Step3	Evaluation (Hz)	1.02	3.39	6.10	9.00	11.87	14.07
Step4	Evaluation (Hz)	1.10	3.62	6.16	9.00	11.83	14.08
Step5	Evaluation (Hz)	1.04	3.49	6.11	8.98	11.90	14.20

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- Step 1: consider only two model parameters (column rotational stiffness at the support and the same beam rotational stiffness for each floor).
- Step 2: update Step 1 using seven model parameters (column rotational stiffness at the support and different beam rotational stiffness for each floor).
- Step 3: update Step 2 using nine model parameters (in addition to the rotational stiffness, the same moment of inertia for column and beam, $\hat{I}_c$ and $\hat{I}_b$, are added).
- Step 4: update Step 3 using 19 model parameters (in addition to rotational stiffness, different moment of inertias for beams and columns are added).
- Step 5: update Step 4 using 25 model parameters (in addition to 19 model parameters in Step 4, six modal damping ratios are included).

From Steps 1 to 4, Eq. (30) is used as the objective function for optimization, whereas Eq. (33) is used as the objective function in Step 5. Sequentially updating the number of model parameters in the FE model will improve the accuracy of the prediction for structural responses. Based on the large-scale nonlinear least-square optimization algorithm, as described in the previous section, the model parameters can be successfully estimated. Another reason for using the sequentially updating procedure is to ensure the stability and convergence of the solution during each step of the numerical optimization process. Since the number of unknown model parameters is large and since the objective function to be optimized is nonlinear, a single-step updating procedure may result in a numerical convergence problem. Table 5 shows the comparison of the identified modal frequencies at different steps, including the results from the SI method, the initial FE model, and the results from Steps 1–5 model updating, for the structure without damage. The identified modal frequencies from the SI method are estimated directly from the experimental data, which are used as the reference for comparison. From Table 5, it is concluded that the estimated modal frequencies obtained from Step 5 are much closer to the SI results because of less physical constraints on the FE model. To verify the updated FE model in Step 5, Fig. 6 shows the comparison of the estimated and experimental floor acceleration responses for 1st, 4th, and 6th floors. As observed from Fig. 6, the estimated floor accelerations are very close to the experimental data.

Based on the seismic response data obtained from the experimental tests for five damage cases (loosened bolt at different floor levels), the proposed FE-based model updating method was applied to all test cases. Fig. 7 shows the bar chart of the identified model parameters (normalized with respect to the undamaged case) for each damage case. Note that the magnitude of a particular bar in Fig. 7 indicates the reduction of the corresponding model parameter due to damage. As observed from the darkest bars (significant reductions) in Fig. 7, the estimated model parameters are in consistent with the damage scenarios where loosened bolts were originally introduced. It is concluded that the proposed FE-based model updating method can be used for the damage detection.

Damage Detection of the Two-Story RC Frame

The proposed FE-based model updating method will be used to detect the damage of the two-story RC frame structure subject to different levels of earthquake excitations. In this preliminary study, the RC frame will be modeled by a system of linear FE equations of motion and the damage will be qualitatively represented by the reduction of linear stiffness parameters. Currently, nonlinear FE models for RC frames are under investigation.

For the RC frame, each floor beam or column is divided into three subelements in order to detect the sequence of crack appearances (in terms of stiffness degradation of the element). As shown in Fig. 8, a total of 15 model parameters are selected for identification

$$\boldsymbol{\beta}_{III} = \langle k_{\theta,c1}\, k_{\theta,b1}\, k_{\theta,b2}\, I^1_{c,1}\, I^2_{c,1}\, I^3_{c,1}\, I^1_{c,2}\, I^2_{c,2}\, I^3_{c,2}\, I^1_{b,1} I^2_{b,1}\, I^1_{b,2}\, I^2_{b,2}\, \xi_1\, \xi_2 \rangle \tag{35}$$

in which the subscript "c" represents the column, "b" indicates the beam, "θ" indicates the rotation, "1" indicates the first floor, and "2" indicates the second floor. The equations of motion for

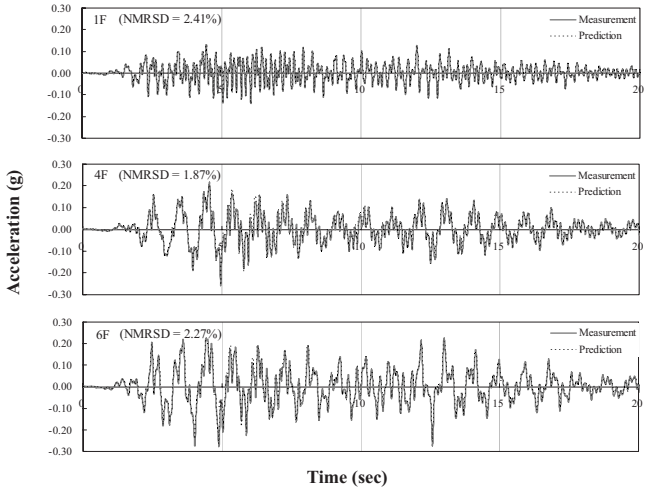


Fig. 6. Comparison between the estimated acceleration responses based on the updated FEM method and the experimental results for 1st, 4th, and 6th floors

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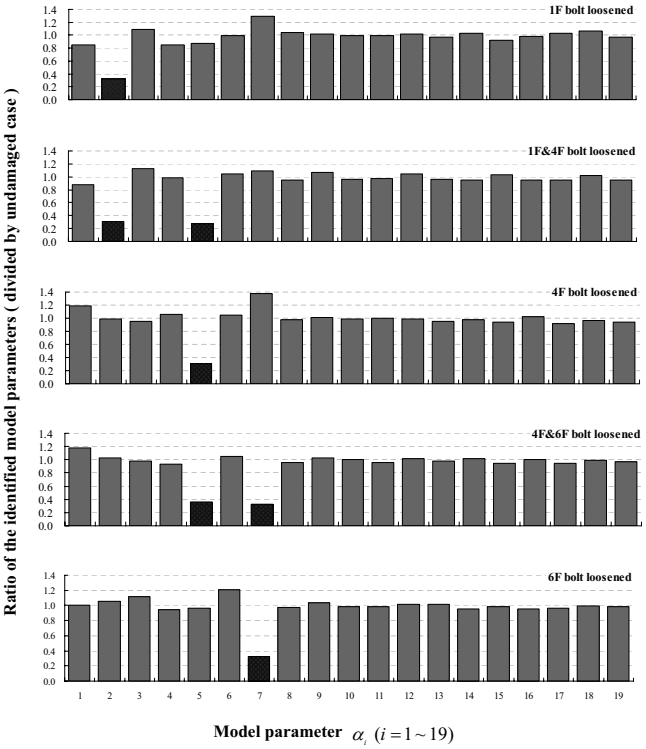


Fig. 7. Ratio of identified model parameters for five different damage cases

the FE model have been established first. This high-order FE model, involving 15 parameters, is then condensed into a two degree-of-freedoms (2DOFs) reduced-order system through the static condensation method resulting in masses m_1 and m_2 as well as modal damping ξ_1 and ξ_2. The reduced-order system is then used for the update of 15 unknowing parameters in Eq. (27) based on the proposed model updating method. It is mentioned that the semirigid connection is also modeled as a zero-length rotational spring at both ends of a beam. Fig. 9 shows the photos of the beam-column joint under different level of excitations. It is clearly observed that the joint damage becomes more and more severe with respect to the level of excitations. Therefore, the modeling of the joint rotational stiffness can provide good damage estimation for the structure.

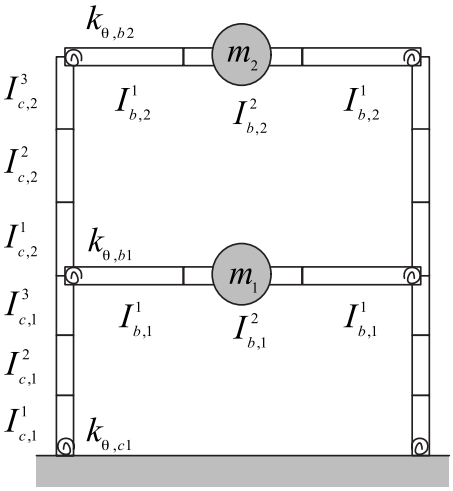


Fig. 8. FE model of the two-story RC frame with 13 model parameters

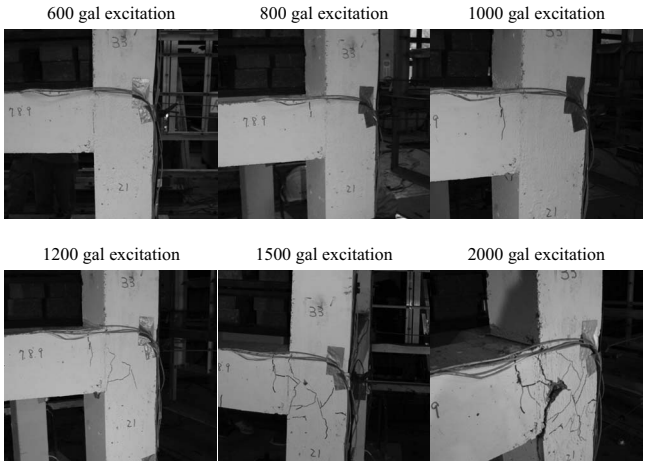


Fig. 9. Photos of the damage scenarios in the beam-column joints of the frame with respect to different levels of seismic excitations

Based on the optimization scheme presented earlier, the moment of inertia of each member and the rotational stiffness of the joints have been identified for different levels of earthquake excitations. Fig. 10 shows the identified moment of inertia of columns and beams as well as the rotational stiffness of the joints of both columns and beams for different levels of excitations. Since the seismic excitations are applied back to back sequentially, the identified model parameters have been degraded with an increasing excitation level. The degradation of the moment of inertia in each column element corresponds to the stiffness reduction caused by the earthquake excitation. Comparisons between the experimental floor accelerations and the estimated acceleration responses obtained from the updated model for different excitation levels are shown in Figs. 11(b–d). Good agreements are observed indicating the accuracy of the proposed FE model updating method. The improvement obtained by the model updating can also be examined from a comparison between the acceleration response predicted by the initial FE model [Fig. 11(a)] and that by the updated FE model [Fig. 11(b)]. It is observed from Fig. 11(b) that the updated FE model can predict the floor acceleration response much better. Based on the identified model parameters, the interstory stiffness can be estimated. Fig. 12 shows the degradation of the interstory stiffness with respect to the input excitation level. To quantify the damage pattern of this frame structure after different levels of seismic excitations, the identified moment of inertia (normalized with respect to the undamaged result) of each element of the frame, $I_{\text{damage}}/I_{\text{undamage}}$, is shown in Fig. 13. Four levels of ratios (damage indices) are defined in Fig. 13, and a smaller value of $I_{\text{damage}}/I_{\text{undamage}}$ indicates a more severe damage. It is interesting to note that in some test cases the identified damage index is higher in the middle part of the beam than at both ends, such as in the cases of 800-gal and 1,200-gal excitations. This does not mean that the damage in the middle element of the beam is subject to a more severe damage than the element at both ends. This is due to the introduction of the joint element, which may smear the damage severity of the beam element. For example, in the case of 1,200-gal excitation, the identified severity of the joint damage is higher than the elements at both ends, which in turn may enhance the damage severity in the middle

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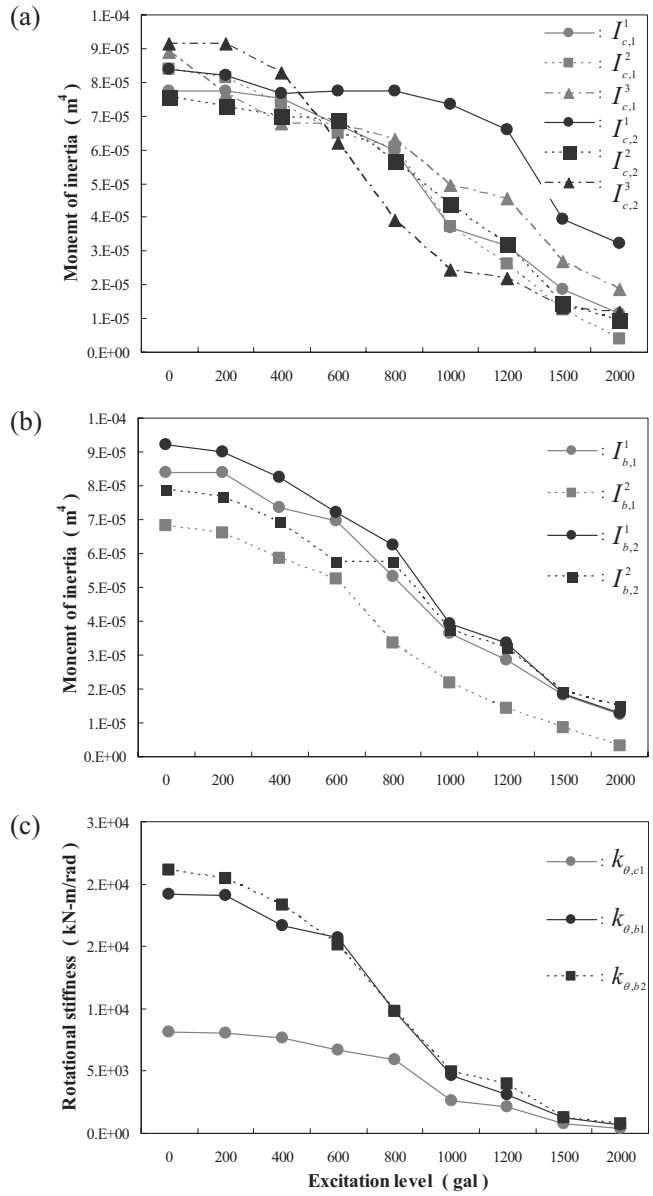


Fig. 10. Identified model parameters after different levels of excitations in PGA: (a) column moment of inertia; (b) beam moment of inertia; and (c) rotational stiffness of joints

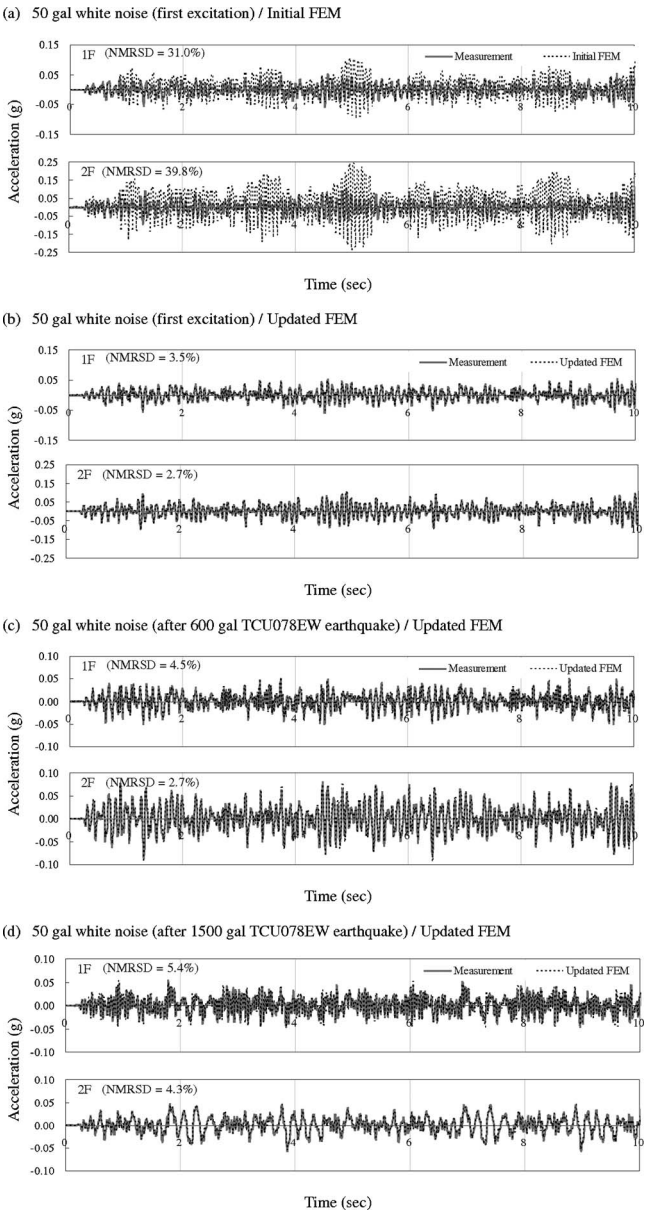


Fig. 11. Comparison between the floor acceleration responses predicted by the initial FE model and the updated model for 50 gal of white noise excitations: (a) the initial FE mode; (b) the updated model; (c) the updated model after 600 gal of seismic excitations; and (d) the updated model after 1,500 gal of seismic excitations

element of the beam. From the results presented earlier (Figs. 10–13) it is shown that the proposed FE-based updating method is capable of detecting structural damages.

Conclusions

This paper proposes a systematic procedure to estimate the structural damages from the experimental test data using a FE model updating approach and a system identification technique. For the FE model of a steel frame structure, the connection stiffness of a beam-to-column joint is modeled by a zero-length rotational spring at the end of the beam element. The rotational rigidity is defined by the modification factor of the connection rigid factor. Then, the damage severity of a joint is defined by the reduction ratio of the rotational rigidity. The SI technique is used first to identify the modal properties of the structure. Based on the results obtained from the system identification and the FE model updating technique, the stiffness factors (bending stiffness and rotational stiffness factors of joints) of the flexibly connected beam elements can be identified using a large-scale optimization technique. Verifications of the proposed method have been demonstrated through a shaking table test of a six-story steel frame with controlled joint damages (loosened bolts) and a RC frame subject to different levels of ground excitations. In general, the proposed method can be applied to full-scale building structures if the measurements for the acceleration response of each floor can be obtained. Finally, it is shown that an adequate parameterization of the FE model is essential in order to reflect the true physical behavior of the system characteristics.

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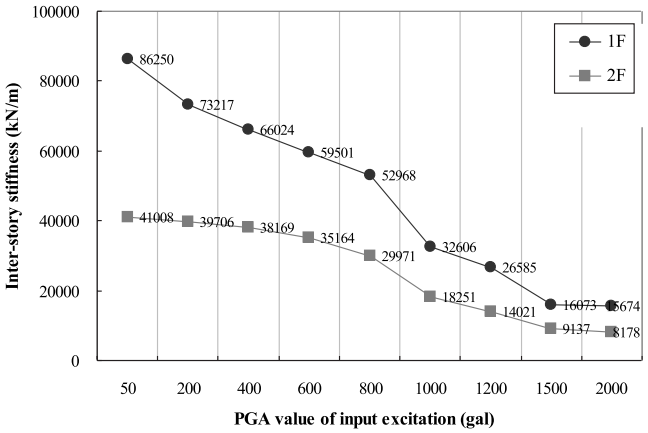


Fig. 12. Identified changes of interstory stiffness after different excitation levels of peak ground acceleration

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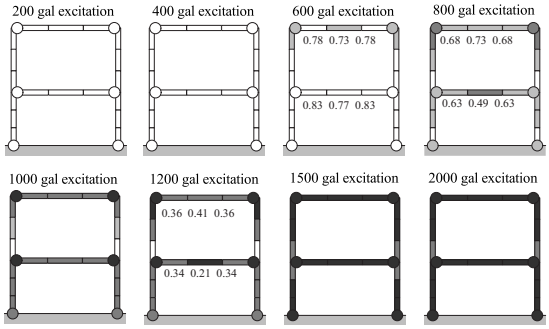


Fig. 13. Eight different damage patterns (each after different levels of excitations) of the two-story RC frame (damage index: 0.75–1.0; 0.5–0.75; 0.25–0.5; 0.0–0.25)

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